

Business Case: Target-SQL

Description:

Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.

This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.

By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

Problem Statement:

Assuming you are a data analyst/ scientist at Target, you have been assigned the task of analyzing the given dataset to extract valuable insights and provide actionable recommendations.

✚ Q1: Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

❖ Q1.1: Data type of all columns in the "customers" table.

➤ SQL Query:

```
SELECT column_name, data_type
FROM `target`.INFORMATION_SCHEMA.COLUMNS
WHERE table_name = 'customers';
```

➤ o/p:

JOB INFORMATION		RESULTS .	JSON	EXECUTION DET
Row	column_name ▼	data_type ▼		
1	customer_id	STRING		
2	customer_unique_id	STRING		
3	customer_zip_code_prefix	INT64		
4	customer_city	STRING		
5	customer_state	STRING		

❖ **Q1.2:** Get the time range between which the orders were placed.

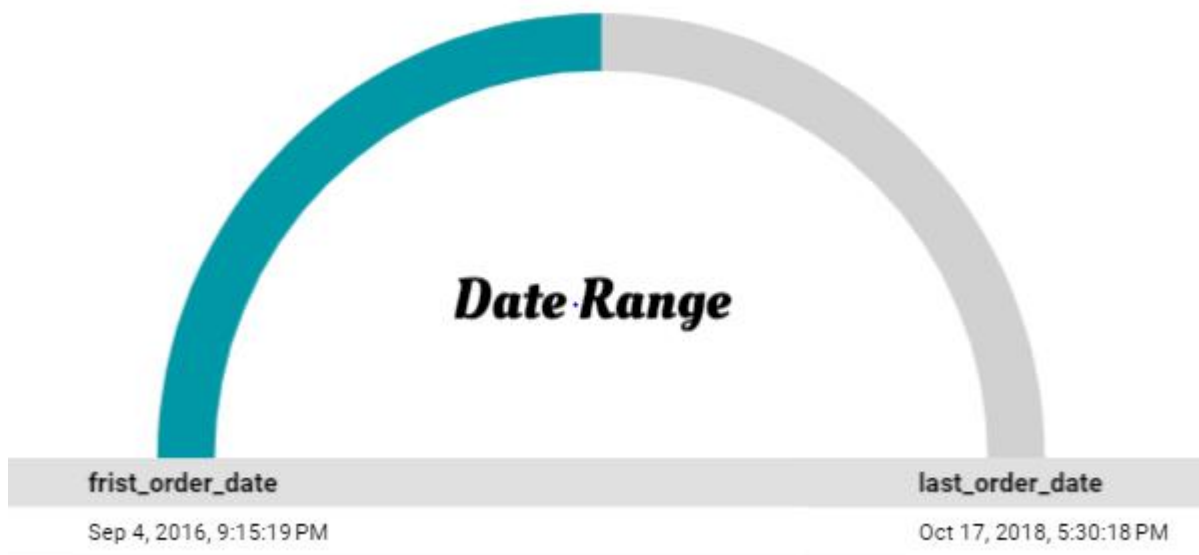
➤ **SQL Query:**

```
SELECT  
MIN(order_purchase_timestamp) AS first_order_date,  
MAX(order_purchase_timestamp) AS last_order_date  
FROM `target.orders`;
```

➤ **o/p:**

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS
Row	first_order_date		last_order_date	
1	2016-09-04 21:15:19 UTC		2018-10-17 17:30:18 UTC	

➤ **Chart:**



❖ **Q1.3:** Count the number of Cities and States in our dataset.

➤ **SQL Query:**

```
with all_city_state as
(
SELECT
geolocation_city AS city,
geolocation_state AS state
FROM `target.geolocation`

UNION DISTINCT

SELECT
customer_city AS city,
customer_state AS state
FROM `target.customers`

UNION DISTINCT

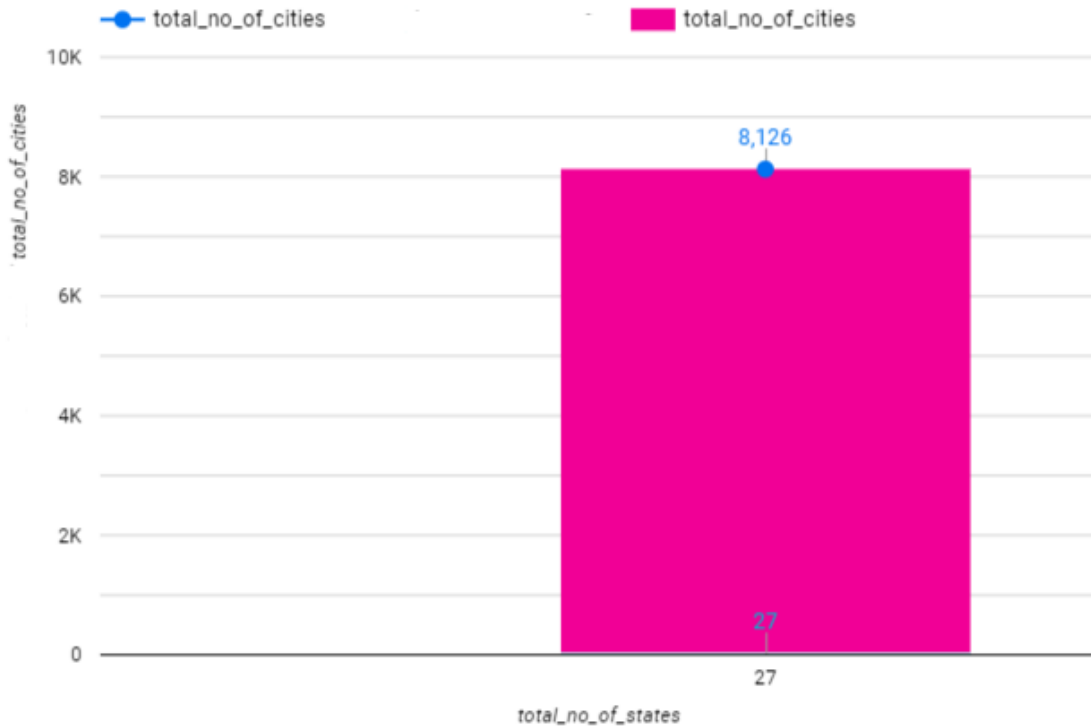
SELECT
seller_city AS city,
seller_state AS state
FROM `target.sellers`
)

SELECT
COUNT(DISTINCT city) as total_no_of_cities,
COUNT(DISTINCT state) as total_no_of_states
FROM all_city_state;
```

➤ **o/p:**

JOB INFORMATION		RESULTS	JSON
Row	total_no_of_cities	total_no_of_states	
1	8126	27	

➤ Chart:



➤ Insights & Recommendations For Q1:

- Customer table has 5 columns to store customer data like customer ID, customer unique ID, zip code, city and state. It will contain new, old all the customer details.
- The data present in the dataset for the time range between which orders were placed is 2016-09-04 21:15:19 UTC to 2018-10-17 17:30:18 UTC.
- There are total 8,126 unique cities and 27 unique states present in the dataset.

🚦 Q2: In-depth Exploration:

❖ Q2.1: Is there a growing trend in the no. of orders placed over the past years?

➤ SQL Query 1:

```
WITH order_growth_year AS
(
SELECT
EXTRACT(year FROM order_purchase_timestamp) AS order_year,
COUNT(order_id) AS no_of_orders,
LAG(COUNT(order_id)) OVER(ORDER BY COUNT(order_id)) AS
next_year_order_count
FROM `target.orders`
GROUP BY order_year
ORDER BY 1
)

SELECT
order_year,
no_of_orders,
ROUND(((no_of_orders -
next_year_order_count)/next_year_order_count)*100,2) AS
growth_percentage_year
FROM order_growth_year
ORDER BY 1;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION TIME
Row	order_year	no_of_orders	growth_percentage_year	
1	2016	329		
2	2017	45101	13608.51	
3	2018	54011	19.76	

➤ SQL Query 2:

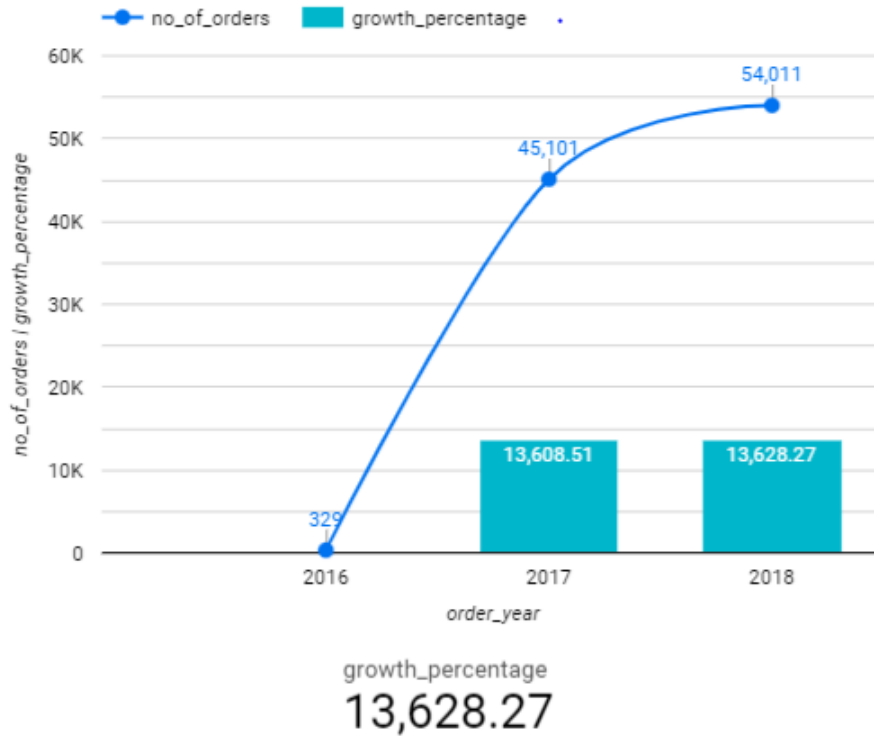
```
WITH order_growth_month_lag AS
(
SELECT
*,
SUM(no_of_orders) OVER(ORDER BY order_year_month) AS
gradually_increasing_total_order_per_year_month,
LAG(no_of_orders) OVER(ORDER BY order_year_month) AS
next_month_order_count
FROM
(
SELECT
FORMAT_TIMESTAMP('%Y-%m', order_purchase_timestamp) AS
order_year_month,
COUNT(order_id) AS no_of_orders,
FROM `target.orders`
GROUP BY order_year_month
ORDER BY 1,2
) AS order_growth_month
ORDER BY 1,2
)

SELECT
order_year_month,
no_of_orders,
ROUND(((no_of_orders -
next_month_order_count)/next_month_order_count)*100,2) AS
growth_percentage_per_month,
gradually_increasing_total_order_per_year_month,
FROM order_growth_month_lag
ORDER BY 1,2;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	order_year_month	no_of_orders	growth_percentage_per_month	gradually_increasing_total_order_per_year_month	
1	2016-09	4	null	4	
2	2016-10	324	8000.0	328	
3	2016-12	1	-99.69	329	
4	2017-01	800	79900.0	1129	
5	2017-02	1780	122.5	2909	
6	2017-03	2682	50.67	5591	
7	2017-04	2404	-10.37	7995	
8	2017-05	3700	53.91	11695	
9	2017-06	3245	-12.3	14940	
10	2017-07	4026	24.07	18966	

➤ Charts:



- ❖ **Q2.2:** Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

➤ **SQL Query:**

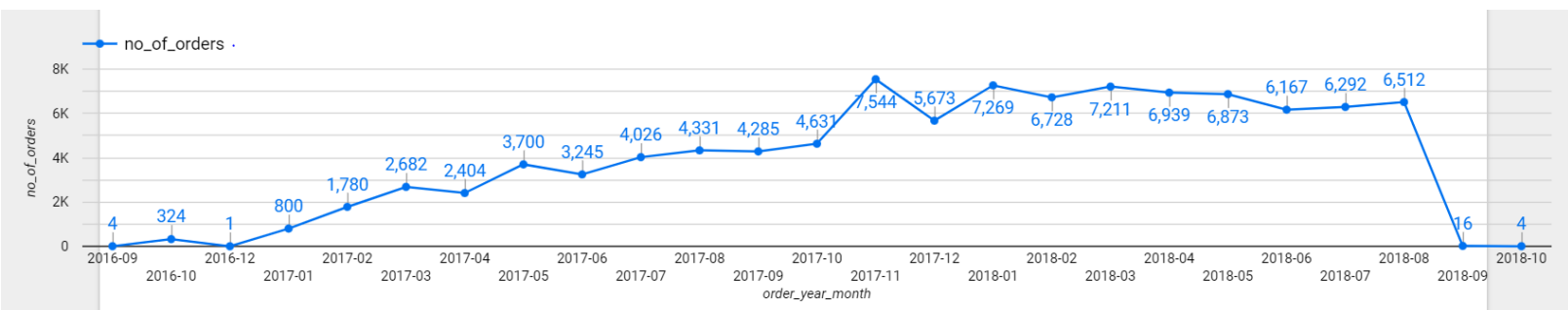
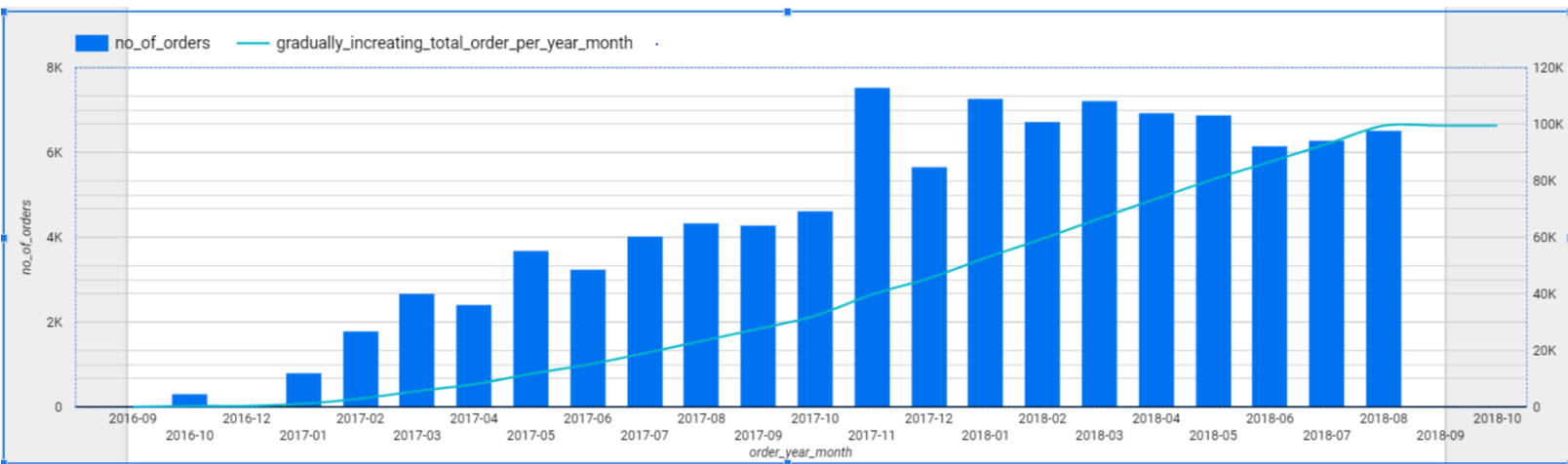
```
WITH order_growth_month_lag AS
(
SELECT
*,
LAG(no_of_orders) OVER(ORDER BY order_year_month) AS
next_month_order_count
FROM
(
SELECT
FORMAT_TIMESTAMP('%Y-%m', order_purchase_timestamp) AS
order_year_month,
EXTRACT(year FROM order_purchase_timestamp) AS order_year,
EXTRACT(month FROM order_purchase_timestamp) AS order_months,
COUNT(order_id) AS no_of_orders
FROM `target.orders`
GROUP BY order_year_month, order_year, order_months
ORDER BY 1,2
) AS order_growth_month
ORDER BY 1,2
)

SELECT
order_year_month,
no_of_orders,
ROUND(((no_of_orders -
next_month_order_count)/next_month_order_count)*100,2) AS
growth_percentage,
FIRST_VALUE(order_months) OVER(PARTITION BY order_year ORDER BY
no_of_orders DESC) AS peak_month
FROM order_growth_month_lag
ORDER BY 1,2;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION TIME
Row	order_year_month	no_of_orders	growth_percentage	peak_month	
1	2016-09	4	null	10	
2	2016-10	324	8000.0	10	
3	2016-12	1	-99.69	10	
4	2017-01	800	79900.0	11	
5	2017-02	1780	122.5	11	
6	2017-03	2682	50.67	11	
7	2017-04	2404	-10.37	11	
8	2017-05	3700	53.91	11	
9	2017-06	3245	-12.3	11	
10	2017-07	4026	24.07	11	

➤ Charts:



- ❖ **Q2.3:** During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
- 0-6 hrs : Dawn
 - 7-12 hrs : Mornings
 - 13-18 hrs : Afternoon
 - 19-23 hrs : Night

➤ **SQL Query:**

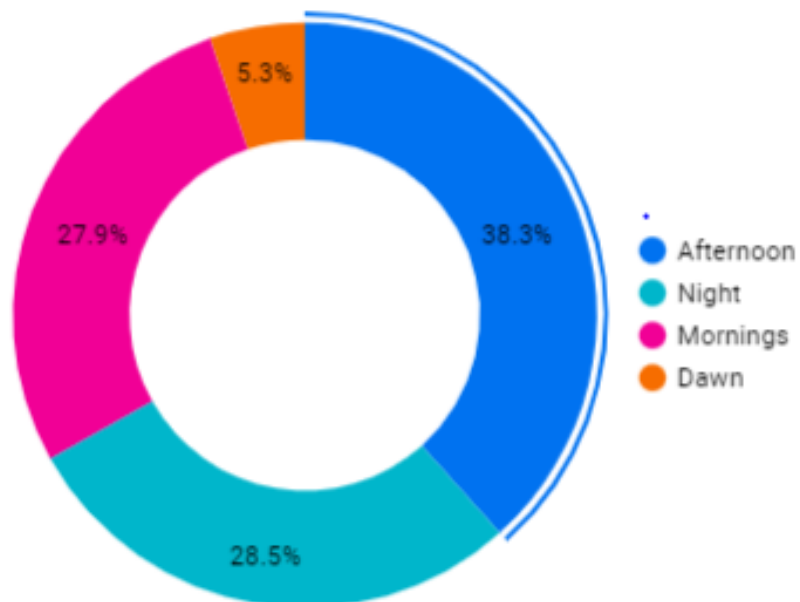
```
WITH order_time_of_day AS
(
SELECT
CASE
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 00 AND 06
    THEN 'Dawn'
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 07 AND 12
    THEN 'Mornings'
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 13 AND 18
    THEN 'Afternoon'
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 19 AND 23
    THEN 'Night'
END AS time_of_day,
COUNT(order_id) no_of_orders
FROM `target.orders`
GROUP BY time_of_day
ORDER BY 2 DESC
)

SELECT
time_of_day,
no_of_orders,
CONCAT(
    (SELECT time_of_day,
    FROM order_time_of_day
    WHERE no_of_orders = (
    SELECT
    MAX(no_of_orders)
    FROM order_time_of_day
    )),', n/o orders: ',MAX(no_of_orders) OVER()
    ) AS most_orders_time
FROM order_time_of_day
ORDER BY 2 DESC;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS
Row	time_of_day	no_of_orders	most_orders_time	
1	Afternoon	38135	Afternoon, n/o orders: 38135	
2	Night	28331	Afternoon, n/o orders: 38135	
3	Mornings	27733	Afternoon, n/o orders: 38135	
4	Dawn	5242	Afternoon, n/o orders: 38135	

➤ Chart:



➤ Insights & Recommendations For Q2:

- There is a substantial growth trend in the number of orders placed over time. The number of orders started with only one in December 2016 and experienced significant growth in subsequent months, reaching a peak of 7,544 orders in November 2017. There is a gradual increase in the number of orders placed each month over the past years. The total number of orders is consistently increasing, indicating a positive growth trend.
- There is general hike in no of order during January due to new year's. And a trend in 2018 due to the FIFA world cup. While there is overall growth, some months experienced a decrease in the number of orders compared to the previous month. For instance, there was a dip in order count from November 2017 to December 2017, with a decrease of 24.8%. These fluctuations suggest variations in customer behavior and demand. These fluctuations may be influenced by factors such as seasonal variations, marketing campaigns, or customer preferences. Monitor the monthly order data to identify long-term growth trends. By understanding the gradual increase in order counts, the company can better allocate resources and plan for future expansion.
- Identify the months with the highest order counts and leverage them for maximum sales and revenue. Allocate additional resources, enhance marketing efforts, and introduce promotions or special offers during these peak periods to capitalize on customer demand, maximize sales and customer engagement. There is evidence of monthly seasonality in the number of orders. Certain months show higher order counts compared to others. For example, January 2017 saw a substantial increase of 800 orders compared to the previous month, indicating a peak period. Similarly, November 2017 had the highest number of orders, reflecting a significant seasonal peak.
- Implement customer engagement strategies to encourage repeat orders and customer loyalty. This can include personalized offers, loyalty programs, or referral incentives. Improving customer satisfaction and building long-term relationships can positively impact order volume. Streamline internal processes and operations to handle increasing order volumes effectively. Identify bottlenecks, automate manual tasks, and enhance supply chain management to ensure timely order fulfillment and customer satisfaction.
- Analyze the months with decreased order counts to understand the underlying reasons. Explore potential factors such as seasonal factors, competition, or customer preferences. Use this information to develop strategies to mitigate the impact of these dips and regain growth. Utilize the historical order data to forecast demand and plan inventory management effectively. Identify the months with high order counts and ensure sufficient stock availability during those periods. Additionally, optimize inventory during low-demand months to avoid overstocking.
- Implement customer retention strategies to encourage repeat orders and loyalty. Offer incentives, discounts, or loyalty programs to encourage customers to choose your company over competitors consistently. Enhance the overall customer experience by focusing on areas such as website usability, shipping times, product quality, and customer support. Positive experiences can lead to increased customer satisfaction, repeat purchases, and positive word-of-mouth referrals.

- The highest number of orders is placed during the afternoon, with a total of 38,135 orders. This suggests that the majority of Brazilian customers prefer to place their orders in the afternoon. The second highest number of orders is placed at night, with 28,331 orders. Mornings follow closely with 27,733 orders, and the lowest number of orders is placed during dawn, with 5,242 orders.
- Since the afternoon is the peak time for order placement, ensure that there are sufficient staff members available to handle the increased volume during this period. Distribute resources efficiently to minimize wait times and provide a seamless customer experience. Focus marketing efforts on promoting and highlighting the company's offerings during the afternoon hours. Consider running targeted promotions or discounts during this time to attract more customers and capitalize on the peak demand.
- Streamline order fulfillment processes and logistics to ensure prompt delivery during the peak afternoon period. Optimize inventory management to meet the increased demand during this time, minimizing any delays or out-of-stock situations. Leverage customer data and insights to personalize and enhance the customer experience during the afternoon. Implement targeted communication and engagement strategies such as personalized recommendations, tailored offers, or loyalty rewards to encourage repeat orders and increase customer satisfaction.
- Consider extending operating hours into the evening to accommodate the high volume of orders placed during the afternoon and night. This allows customers more flexibility in placing orders and can potentially attract new customers who prefer later time slots for shopping. Continuously monitor and analyze order data to identify any shifts or changes in customer ordering patterns. Stay updated with customer preferences and adapt strategies accordingly to stay ahead of the competition and meet evolving customer needs.

🚦 Q3: Evolution of E-commerce orders in the Brazil region:

❖ **Q3.1:** Get the month on month no. of orders placed in each state.

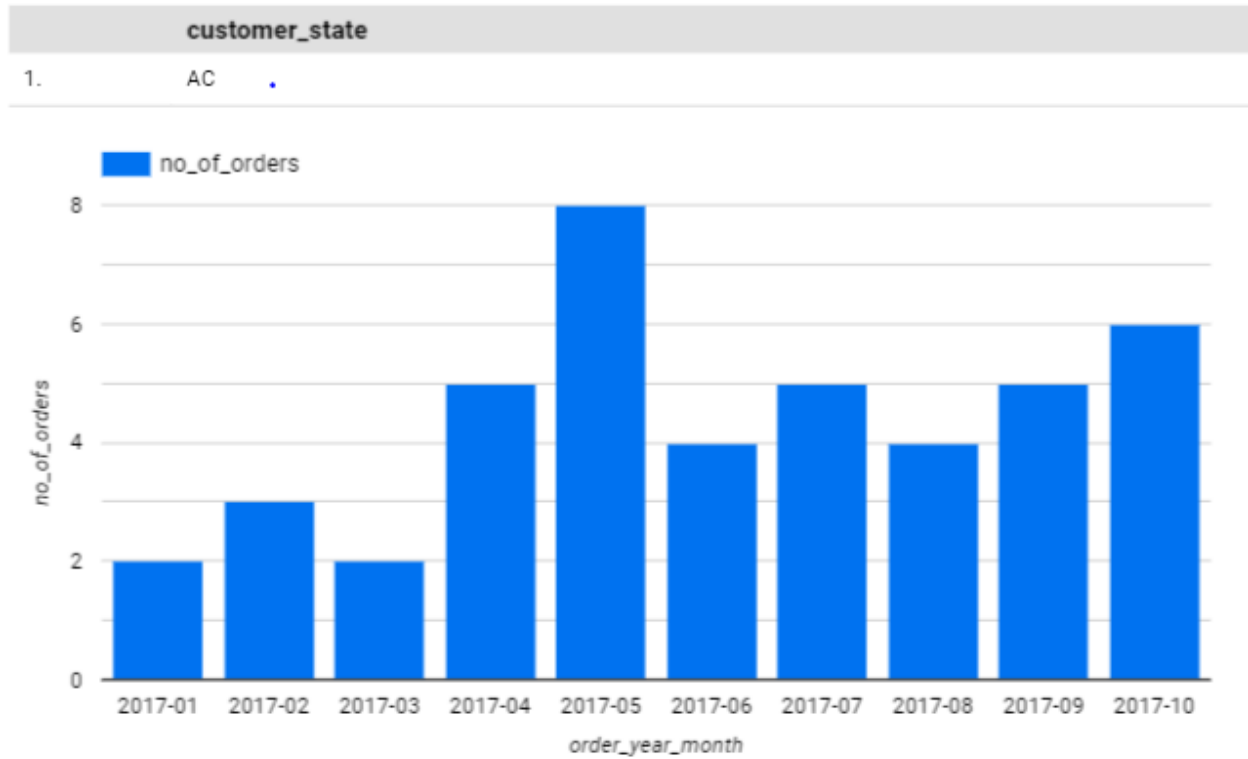
➤ **SQL Query:**

```
SELECT c.customer_state,  
FORMAT_TIMESTAMP('%Y-%m', order_purchase_timestamp) AS  
order_year_month,  
COUNT(o.order_id) no_of_orders  
FROM `target.customers` c  
INNER JOIN `target.orders` o  
ON c.customer_id = o.customer_id  
GROUP BY c.customer_state, order_year_month  
ORDER BY 1,2,3;
```

➤ **o/p:**

JOB INFORMATION		RESULTS	JSON	EX
Row	customer_state	order_year_month	no_of_orders	
1	AC	2017-01	2	
2	AC	2017-02	3	
3	AC	2017-03	2	
4	AC	2017-04	5	
5	AC	2017-05	8	
6	AC	2017-06	4	
7	AC	2017-07	5	
8	AC	2017-08	4	
9	AC	2017-09	5	
10	AC	2017-10	6	

➤ Chart:



❖ Q3.2: How are the customers distributed across all the states?

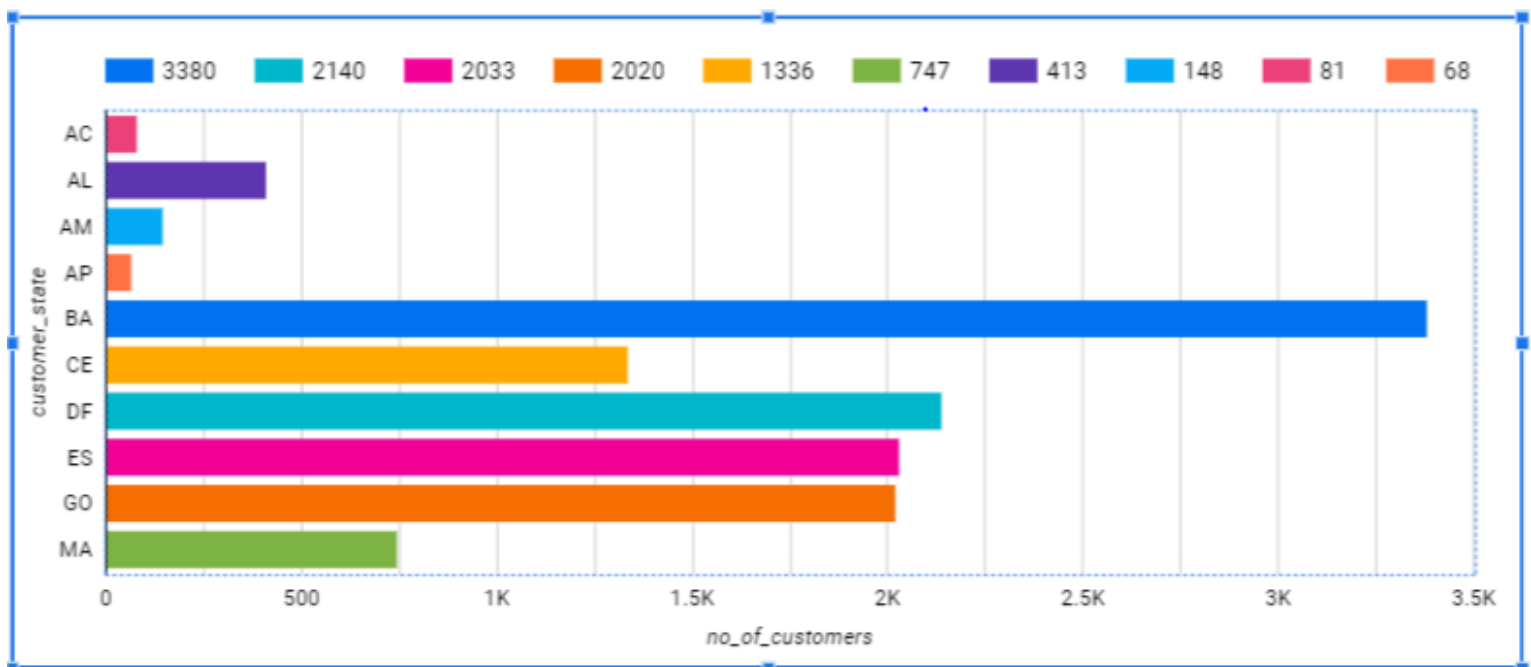
➤ SQL Query:

```
SELECT customer_state,  
COUNT(customer_id) no_of_customers  
FROM `target.customers`  
GROUP BY customer_state  
ORDER BY 1;
```


➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXE
Row	customer_state	no_of_customers		
1	AC	81		
2	AL	413		
3	AM	148		
4	AP	68		
5	BA	3380		
6	CE	1336		
7	DF	2140		
8	ES	2033		
9	GO	2020		
10	MA	747		

➤ Chart:



➤ Insights & Recommendations For Q3:

- e.g: MA. In February 2017, there were 11 orders placed in MA. The number of orders gradually increased from February 2017 to November 2017, reaching a peak of 56 orders in November 2017. After November 2017, the number of orders started to decrease, reaching 46 orders in April 2018.
- Implement strategies to retain customers and encourage repeat orders. Offer personalized recommendations, loyalty programs, or exclusive discounts to incentivize customers in different states to continue placing orders. Collect feedback from customers in states to understand their preferences and identify areas for improvement. Use this feedback to enhance the ordering experience, address any pain points, and ensure customer satisfaction.
- Increase marketing efforts in different states to attract new customers and increase order volume. Utilize targeted advertisements, social media campaigns, or partnerships with local influencers to raise awareness about the company's offerings and encourage more orders. Assess the demand and preferences of customers in states and consider expanding the product range to cater to their specific needs. Conduct market research to identify potential product categories or variations that could attract more orders.
- Tailor marketing campaigns and advertisements specific to each state to attract new customers. Consider regional preferences, cultural nuances, and local trends to effectively engage with potential customers. Identify states with lower customer counts and focus on expanding the company's presence in these regions. Increase marketing efforts, establish partnerships with local businesses, or offer incentives to encourage more customers to join.
- Customize the shopping experience for customers in different states. Consider offering region-specific products, services, or promotions that align with local tastes and preferences. Provide excellent customer support services to address any inquiries or concerns promptly. Ensure clear communication channels and efficient problem resolution to build trust and loyalty among customers.

✚ Q4: Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

- ❖ **Q4.1:** Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

➤ **SQL Query:**

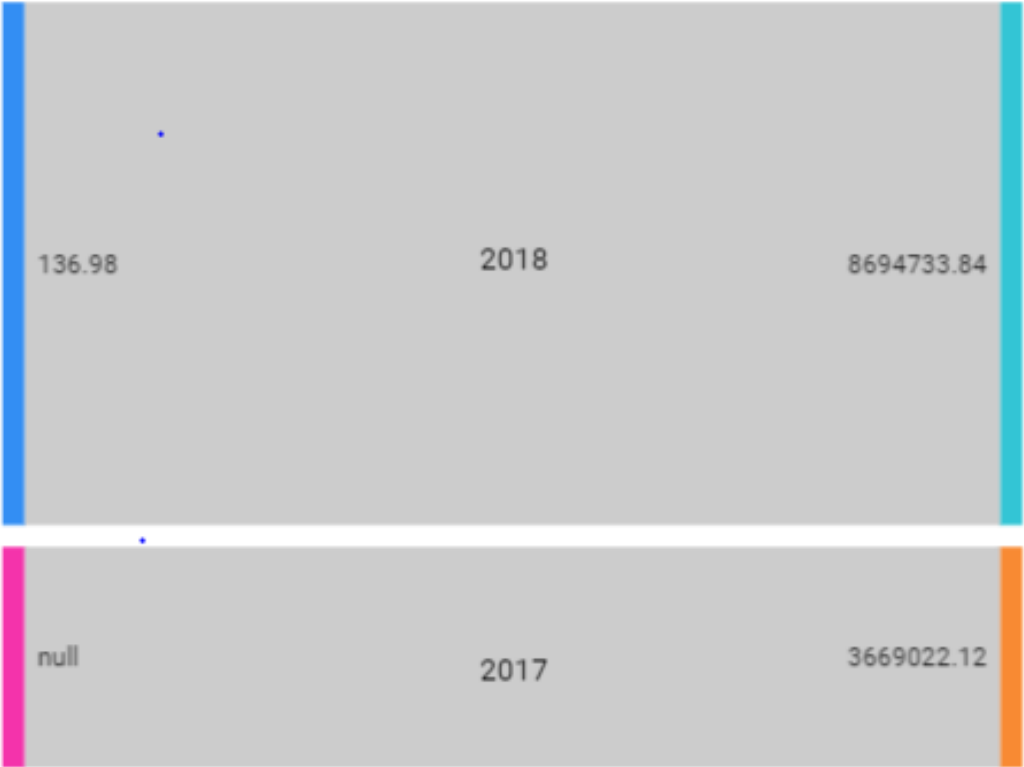
```
WITH order_cost_year_month AS
(
  SELECT *,
  LAG(total_order_value) OVER(ORDER BY order_year)
  next_month_total_order_value
  FROM
  (
    SELECT
    EXTRACT(year FROM o.order_purchase_timestamp) order_year,
    ROUND(SUM(p.payment_value),2) total_order_value,
    FROM `target.orders` o
    INNER JOIN `target.payments` p
    ON o.order_id = p.order_id
    WHERE (EXTRACT(year FROM o.order_purchase_timestamp) BETWEEN 2017 AND
    2018) AND
    (EXTRACT(month FROM o.order_purchase_timestamp) BETWEEN 1 AND 8)
    GROUP By order_year
    ORDER BY 1,2
  )
  ORDER BY order_year
)

SELECT
order_year order_jan_aug_year,
total_order_value,
ROUND(((total_order_value -
next_month_total_order_value)/next_month_total_order_value)*100,2)
growth_percentage
FROM order_cost_year_month
ORDER BY 1,2,3;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION
Row	order_jan_aug_year	total_order_value	growth_percentage	
1	2017	3669022.12	null	
2	2018	8694733.84	136.98	

➤ Chart:



❖ **Q4.2:** Calculate the Total & Average value of order price for each state.

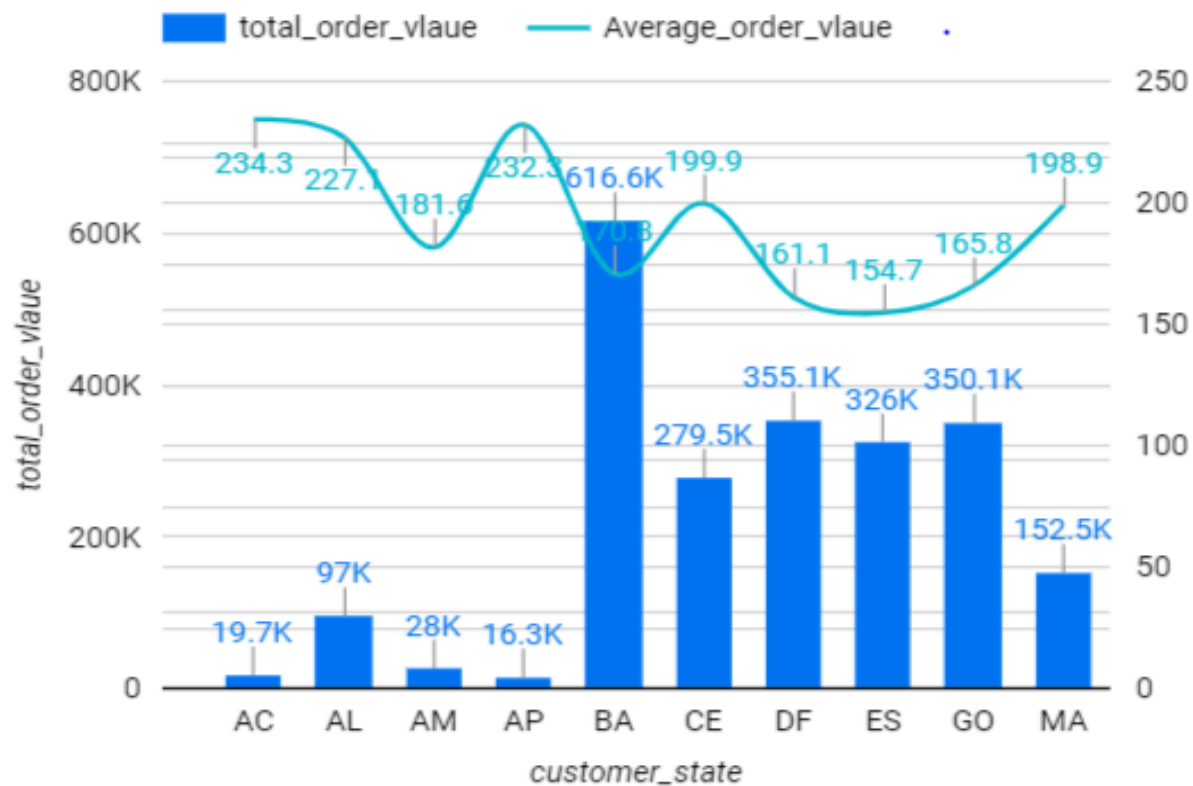
➤ **SQL Query:**

```
SELECT
c.customer_state,
ROUND(SUM(p.payment_value),2) total_order_vlaue,
ROUND(AVG(p.payment_value),2) Average_order_vlaue,
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
INNER JOIN `target.payments` p
ON o.order_id = p.order_id
GROUP BY c.customer_state
ORDER BY 1;
```

➤ **o/p:**

JOB INFORMATION		RESULTS	JSON	EXECUTI
Row	customer_state	total_order_vlaue	Average_order_vlaue	
1	AC	19680.62	234.29	
2	AL	96962.06	227.08	
3	AM	27966.93	181.6	
4	AP	16262.8	232.33	
5	BA	616645.82	170.82	
6	CE	279464.03	199.9	
7	DF	355141.08	161.13	
8	ES	325967.55	154.71	
9	GO	350092.31	165.76	
10	MA	152523.02	198.86	

➤ Chart:



❖ **Q4.3:** Calculate the Total & Average value of order freight for each state.

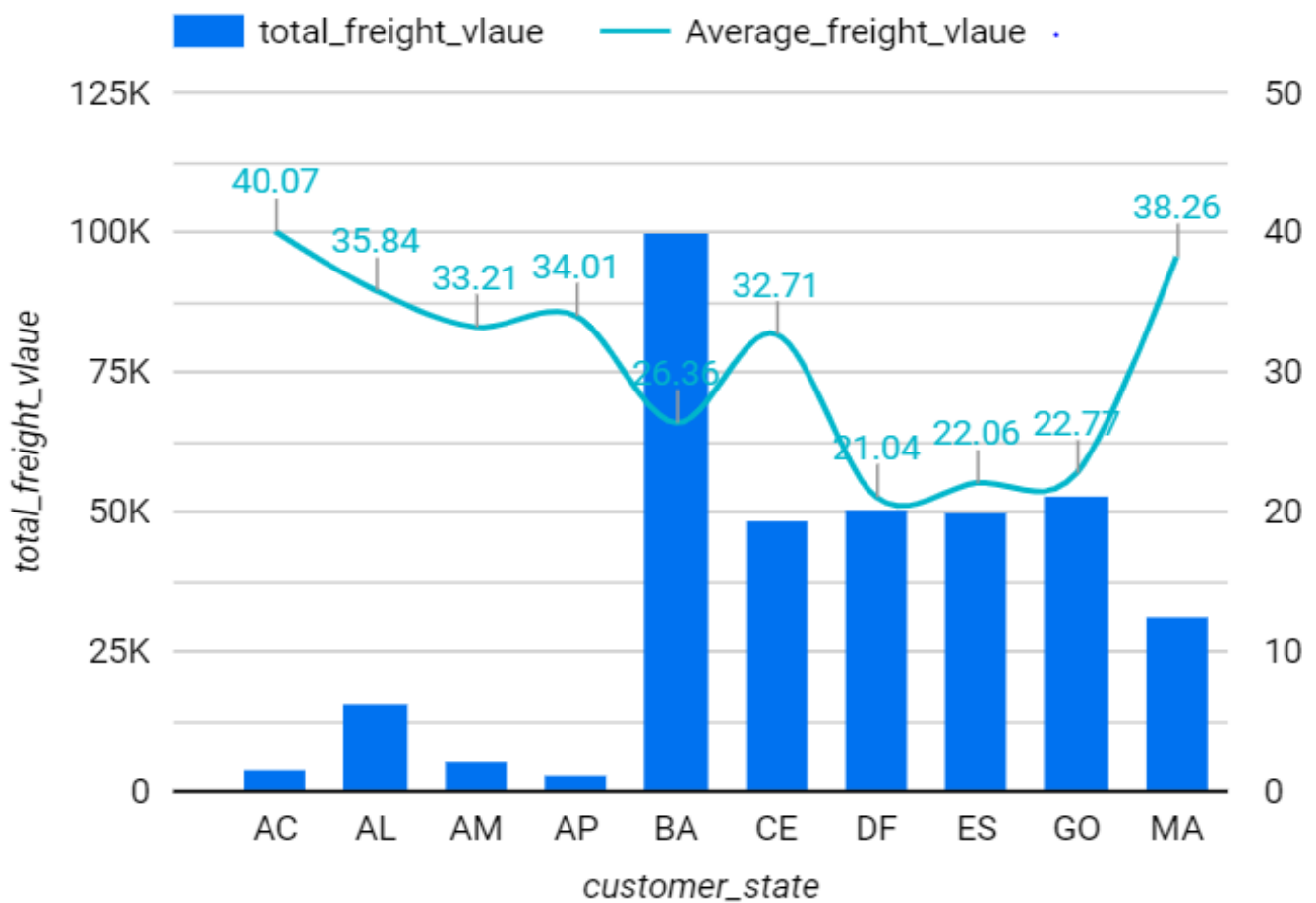
➤ SQL Query:

```
SELECT
c.customer_state,
ROUND(SUM(ot.freight_value),2) total_freight_vlaue,
ROUND(AVG(ot.freight_value),2) Average_freight_vlaue,
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
INNER JOIN `target.order_items` ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
ORDER BY 1;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION D
Row	customer_state	total_freight_vlaue	Average_freight_vlaue	
1	AC	3686.75	40.07	
2	AL	15914.59	35.84	
3	AM	5478.89	33.21	
4	AP	2788.5	34.01	
5	BA	100156.68	26.36	
6	CE	48351.59	32.71	
7	DF	50625.5	21.04	
8	ES	49764.6	22.06	
9	GO	53114.98	22.77	
10	MA	31523.77	38.26	

➤ Chart:



➤ Insights & Recommendations For Q4:

- There has been a significant increase in the cost of orders from 2017 to 2018 (January to August), with a growth percentage of 136.98%. This growth indicates a positive trend in the company's revenue and business performance during the specified period. The company should leverage this positive trend by investing in marketing and promotional activities to sustain and further boost the increase in order value. With increased revenue, allocate resources to improve various aspects of the customer experience, including website usability, product quality, delivery speed, and customer support.
- Evaluate the pricing structure to ensure it aligns with market trends and customer expectations. Conduct regular price analysis and competitor benchmarking to optimize pricing strategies for maximizing revenue. Implement initiatives to enhance customer loyalty and retention. This can include loyalty programs, personalized offers, and targeted marketing campaigns to engage and reward existing customers.
- Assess customer preferences and market demand to identify opportunities for expanding the product catalog. Introduce new product lines or diversify existing offerings to attract a wider customer base and drive additional sales. As order volume increases, focus on streamlining internal processes, supply chain management, and logistics to handle the growing demand effectively. Identify areas for automation or optimization to improve efficiency and reduce costs.
- There is variation in the total and average order values across different states. States like BA, CE, DF, and GO have relatively higher total order values, indicating a potentially larger customer base or higher-priced products in these regions. States like AM, AP, and ES have lower average order values, which could be influenced by factors such as product preferences, local economy, or pricing strategies. The average order value can provide insights into customer spending patterns and preferences in different states.
- Analyze the market and customer behavior in each state to identify specific factors influencing order values. This can include local preferences, purchasing power, competition, and cultural aspects. Segment customers based on geographical regions to better understand their needs and preferences. Tailor marketing strategies and product offerings to target specific regions and customer segments effectively.
- Evaluate pricing strategies and competitive landscape in states with lower average order values. Consider adjusting prices, offering discounts or promotions, or introducing product bundles to increase the perceived value for customers. Implement strategies to encourage customers to increase their order value through upselling and cross-selling techniques.
- Enhance the overall customer experience to increase customer satisfaction and potentially drive higher order values. Focus on aspects such as website usability, product quality, customer support, and delivery speed. Develop targeted marketing campaigns for specific states to create awareness, increase engagement, and drive higher order values. Consider leveraging local events, holidays, or cultural aspects to connect with customers on a deeper level.

- There is variation in the total and average freight values across different states. States like BA, CE, and MA have relatively higher total freight values, indicating potentially larger order volumes or higher shipping costs in these regions. States like DF and MG have lower average freight values, which could be influenced by factors such as distance from distribution centers or logistics efficiency. The average freight value can provide insights into the shipping costs associated with orders in different states.
- Evaluate shipping providers, negotiate better rates, and optimize shipping routes to reduce freight costs. Implement shipping strategies such as bulk shipping, optimizing packaging sizes, or exploring alternative transportation options to lower shipping expenses. Analyze the distribution network and consider establishing regional distribution centers in states with higher order volumes. This can help reduce shipping distances, lower freight costs, and improve delivery times.
- Evaluate the possibility of implementing free shipping thresholds based on order values for specific states. This can incentivize customers to increase their order values and potentially absorb some shipping costs. Improve tracking and visibility of shipments to provide customers with real-time updates on their orders. This can enhance customer satisfaction and reduce inquiries or concerns related to shipping. Collaborate with shipping partners to optimize their operations, improve service levels, and explore options for reduced shipping costs or faster delivery times.

✚ Q5: Analysis based on sales, freight and delivery time.

- ❖ **Q5.1:** Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp
- **diff_estimated_delivery** = order_estimated_delivery_date - order_delivered_customer_date

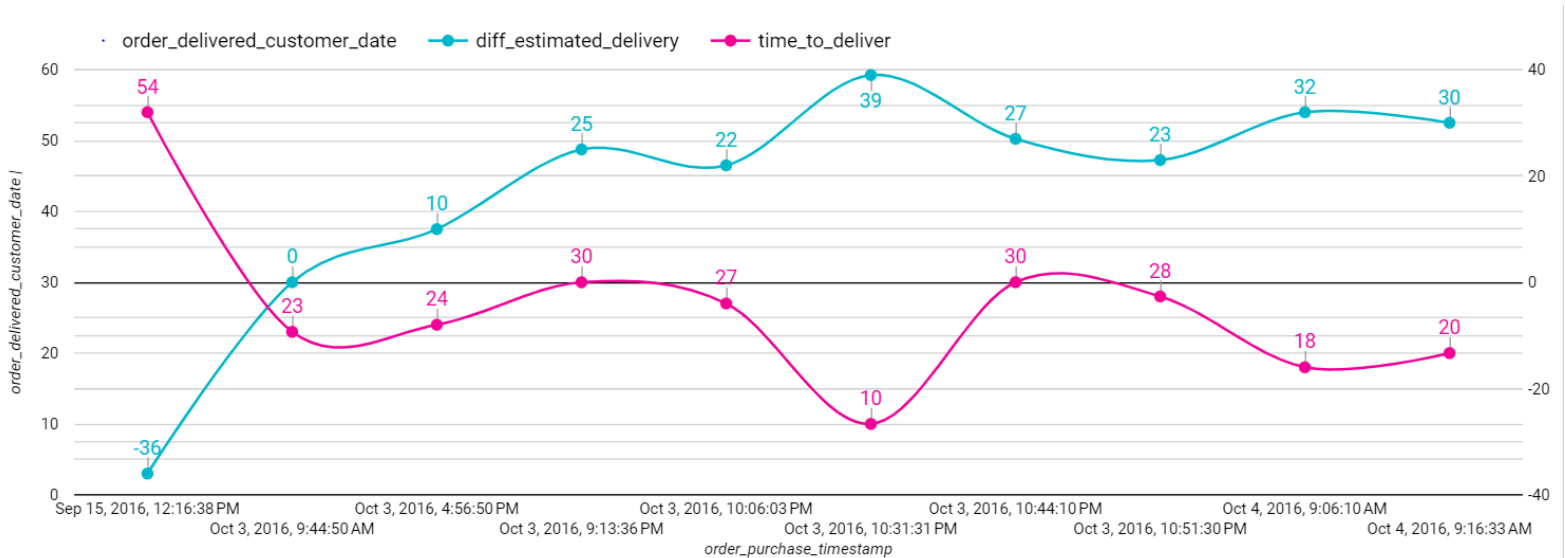
➤ SQL Query:

```
SELECT
order_id,
order_purchase_timestamp,
order_estimated_delivery_date,
order_delivered_customer_date,
DATE_DIFF(order_delivered_customer_date,
order_purchase_timestamp, day) time_to_deliver,
DATE_DIFF(order_estimated_delivery_date,
order_delivered_customer_date, day) diff_estimated_delivery,
FROM
`target.orders`
WHERE LOWER(order_status) = 'delivered'
ORDER BY 2;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH	
Row	order_id	order_purchase_timestamp	order_estimated_delivery_date	order_delivered_customer_date	time_to_deliver	diff_estimated_delivery
1	bfb0f9bdef...	2016-09-15 12:16:38 UTC	2016-10-04 00:00:00 UTC	2016-11-09 07:47:38 UTC	54	-36
2	3b697a20d9...	2016-10-03 09:44:50 UTC	2016-10-27 00:00:00 UTC	2016-10-26 14:02:13 UTC	23	0
3	be5bc2f0da...	2016-10-03 16:56:50 UTC	2016-11-07 00:00:00 UTC	2016-10-27 18:19:38 UTC	24	10
4	a41c8759fb...	2016-10-03 21:13:36 UTC	2016-11-29 00:00:00 UTC	2016-11-03 10:58:07 UTC	30	25
5	d207cc2726...	2016-10-03 22:06:03 UTC	2016-11-23 00:00:00 UTC	2016-10-31 11:07:42 UTC	27	22
6	cd3b8574c8...	2016-10-03 22:31:31 UTC	2016-11-23 00:00:00 UTC	2016-10-14 16:08:00 UTC	10	39
7	ae8a60e4b0...	2016-10-03 22:44:10 UTC	2016-12-01 00:00:00 UTC	2016-11-03 14:04:50 UTC	30	27
8	ef1b29b591...	2016-10-03 22:51:30 UTC	2016-11-25 00:00:00 UTC	2016-11-01 15:14:45 UTC	28	23
9	0a0837a5ee...	2016-10-04 09:06:10 UTC	2016-11-24 00:00:00 UTC	2016-10-22 14:51:18 UTC	18	32
10	1ff217aa612...	2016-10-04 09:16:33 UTC	2016-11-24 00:00:00 UTC	2016-10-24 16:33:45 UTC	20	30

➤ Chart:



❖ **Q5.2:** Find out the top 5 states with the highest & lowest average freight value.

➤ **SQL Query:**

```
WITH low_average_freight_vlaue AS
(SELECT *
FROM
(
SELECT *,
DENSE_RANK() OVER(ORDER BY lowest_average_freight_vlaue) as denserank,
ROW_NUMBER() OVER(ORDER BY lowest_average_freight_vlaue) as row_num
FROM
(
SELECT
c.customer_state,
ROUND(AVG(ot.freight_value),2) lowest_average_freight_vlaue
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
INNER JOIN `target.order_items` ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
ORDER BY 2 DESC
)
ORDER BY lowest_average_freight_vlaue
) WHERE denserank <= 5
)

,

high_average_freight_vlaue AS
(
SELECT *
FROM
(
SELECT *,
DENSE_RANK() OVER(ORDER BY highest_average_freight_vlaue DESC) as
denserank,
ROW_NUMBER() OVER(ORDER BY highest_average_freight_vlaue DESC) as
row_num
FROM
(
SELECT
c.customer_state,
ROUND(AVG(ot.freight_value),2) highest_average_freight_vlaue
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
INNER JOIN `target.order_items` ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
```

```

ORDER BY 2 DESC
)
ORDER BY highest_average_freight_vlaue DESC
) WHERE denserank <= 5
)

```

```

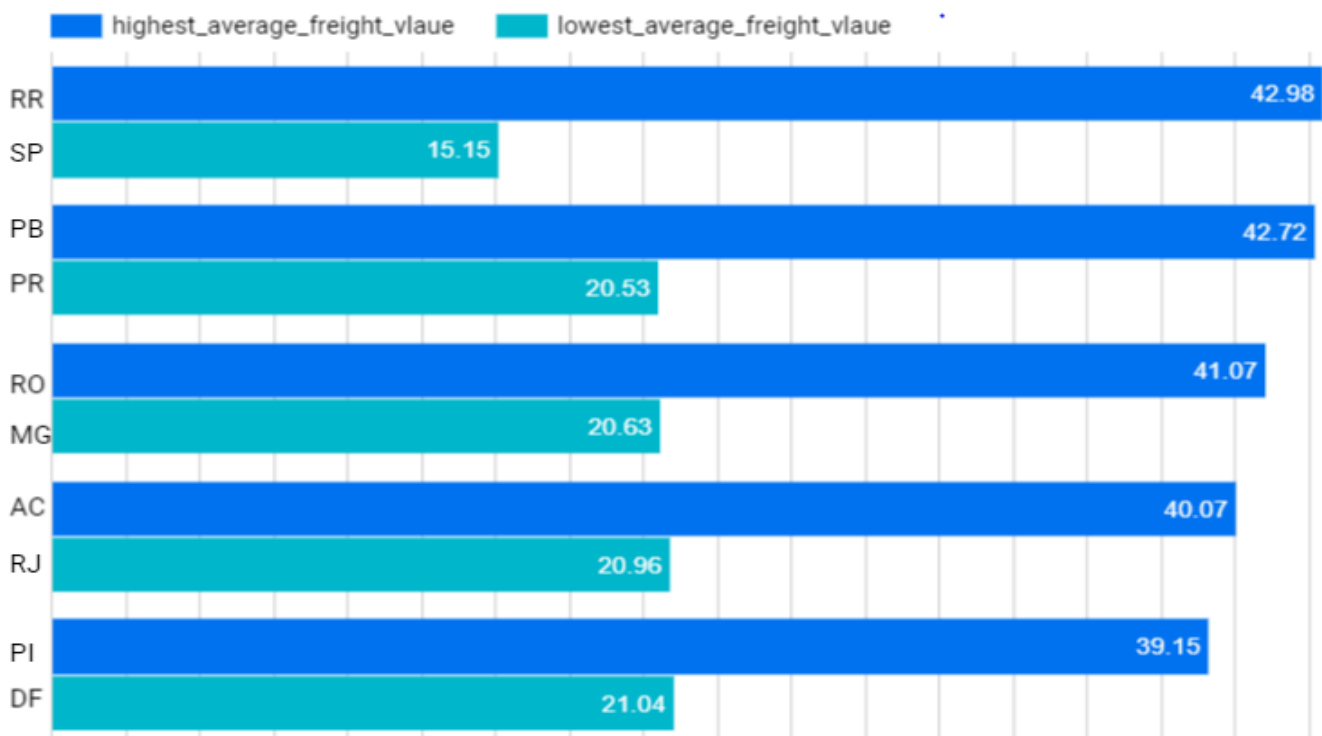
SELECT
h.customer_state AS high_avg_state,
h.highest_average_freight_vlaue,
l.customer_state AS low_avg_state,
l.lowest_average_freight_vlaue
FROM
low_average_freight_vlaue AS l
INNER JOIN high_average_freight_vlaue AS h
ON l.row_num = h.row_num
ORDER BY 2 DESC;

```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	high_avg_state	highest_average_freight_vlaue	low_avg_state	lowest_average_freight_vlaue	
1	RR	42.98	SP	15.15	
2	PB	42.72	PR	20.53	
3	RO	41.07	MG	20.63	
4	AC	40.07	RJ	20.96	
5	PI	39.15	DF	21.04	

➤ Chart:



❖ **Q5.3:** Find out the top 5 states with the highest & lowest average delivery time.

➤ **SQL Query:**

```
WITH low_average_delivery AS
(
  SELECT *
  FROM
  (
    SELECT *,
    DENSE_RANK() OVER(ORDER BY lowest_average_delivery) as denserank,
    ROW_NUMBER() OVER(ORDER BY lowest_average_delivery) as row_num
    FROM
    (
      SELECT
      c.customer_state,
      ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date,
      o.order_purchase_timestamp, day)),2) lowest_average_delivery,
      FROM
      `target.customers` c
      INNER JOIN `target.orders` o
      ON c.customer_id = o.customer_id
      GROUP BY c.customer_state
      ORDER BY 2 DESC
    )
    ORDER BY lowest_average_delivery
  ) WHERE denserank <= 5
)

,
high_average_delivery AS
(
  SELECT *
  FROM
  (
    SELECT *,
    DENSE_RANK() OVER(ORDER BY highest_average_delivery DESC) as denserank,
    ROW_NUMBER() OVER(ORDER BY highest_average_delivery DESC) as row_num
    FROM
    (
      SELECT
      c.customer_state,
      ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date,
      o.order_purchase_timestamp, day)),2) highest_average_delivery,
      FROM
      `target.customers` c
      INNER JOIN `target.orders` o
      ON c.customer_id = o.customer_id
      GROUP BY c.customer_state
      ORDER BY 2 DESC
    )
    ORDER BY highest_average_delivery DESC
  )
)
```

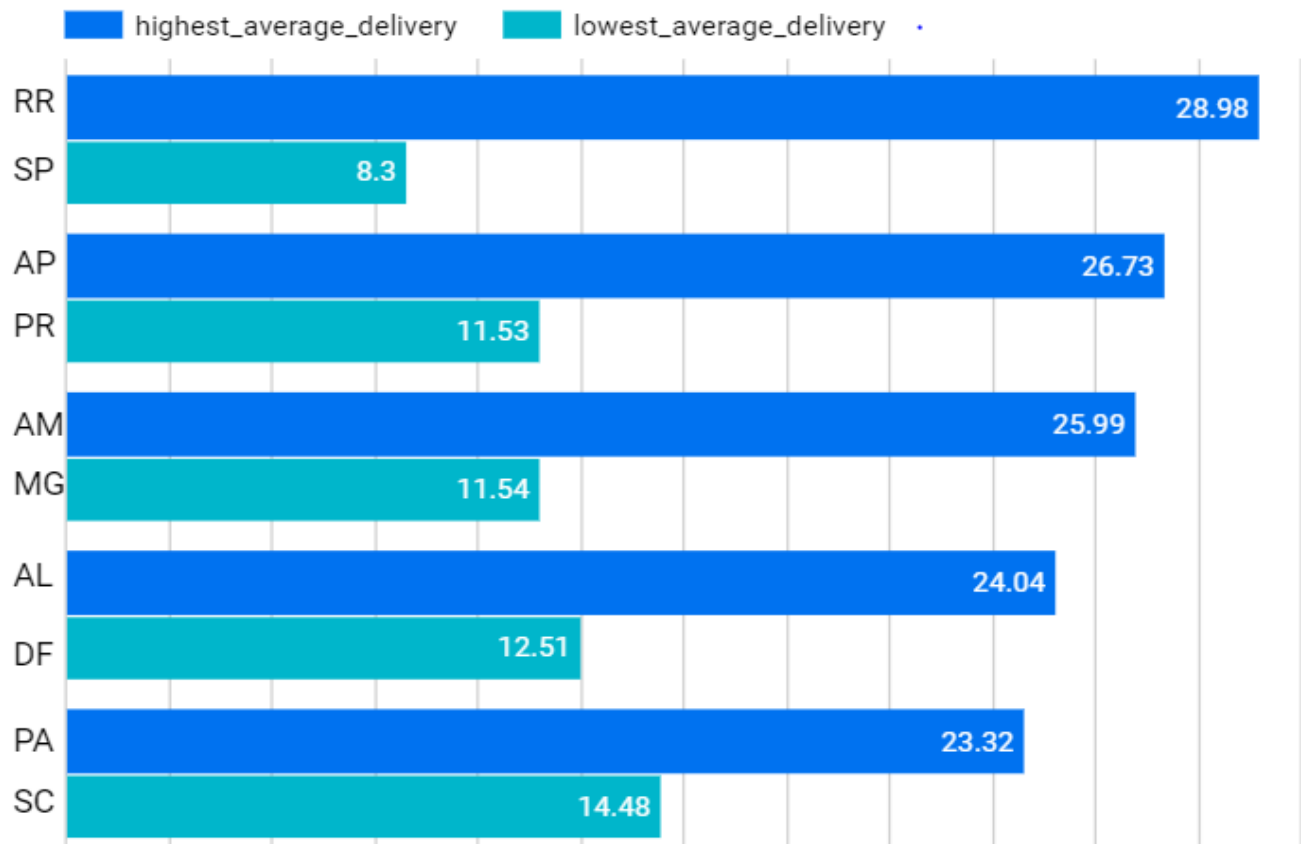
```
) WHERE denserank <= 5
)
```

```
SELECT
h.customer_state AS high_avg_state,
h.highest_average_delivery,
l.customer_state AS low_avg_state,
l.lowest_average_delivery
FROM low_average_delivery AS l
INNER JOIN high_average_delivery AS h
ON l.row_num = h.row_num
ORDER BY 2 DESC;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS		EXECUTIC
Row	high_avg_state	highest_average_delivery	low_avg_state	lowest_average_delivery		
1	RR	28.98	SP	8.3		
2	AP	26.73	PR	11.53		
3	AM	25.99	MG	11.54		
4	AL	24.04	DF	12.51		
5	PA	23.32	SC	14.48		

➤ Chart:



- ❖ **Q5.4:** Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.
You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

➤ **SQL Query:**

```
WITH state_average_delivery AS
(
SELECT
c.customer_state,
ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date,
o.order_delivered_customer_date, day)),2) average_delivery,
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
WHERE LOWER(o.order_status) = 'delivered'
GROUP BY c.customer_state
ORDER BY 2 DESC
)

,

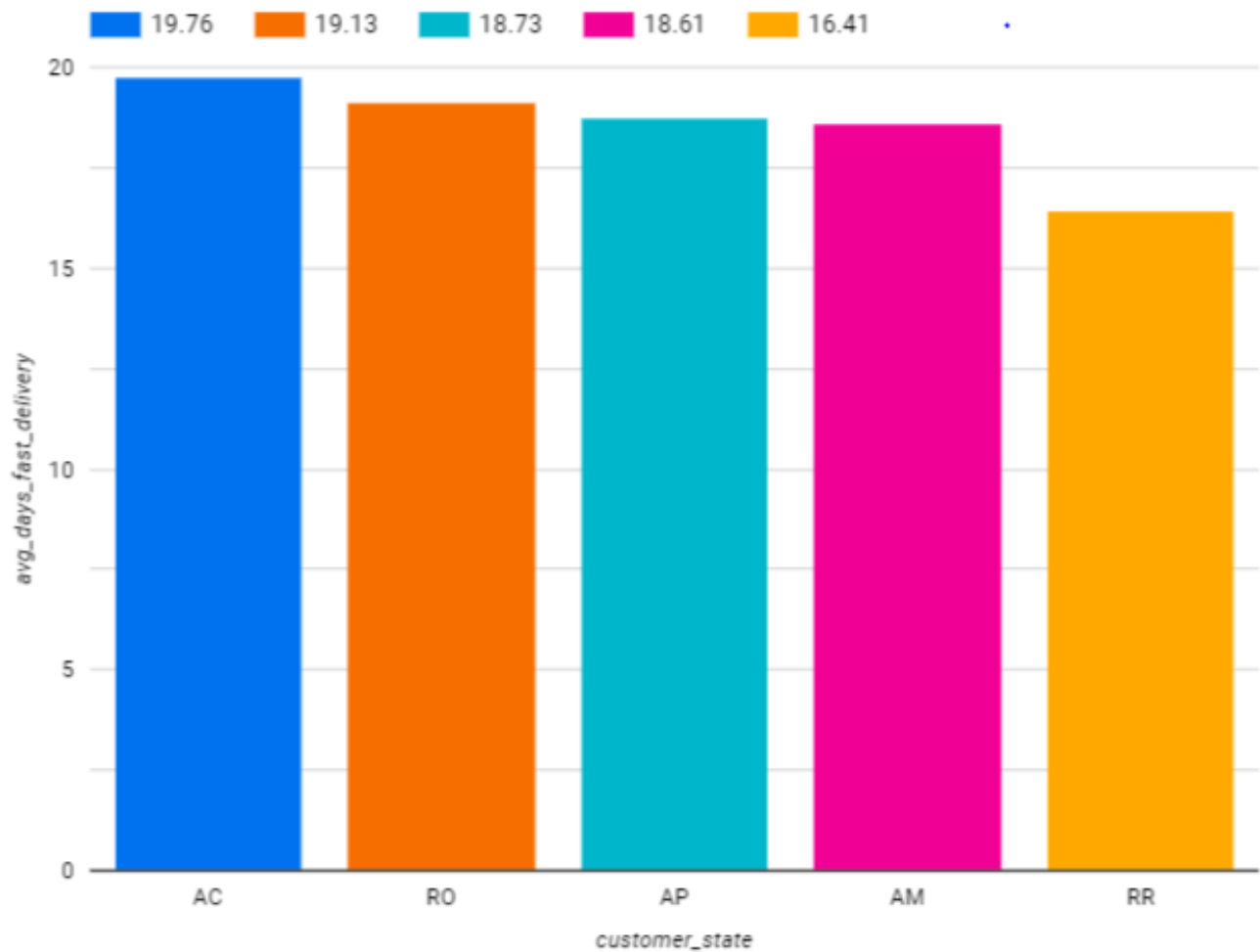
drank AS
(
SELECT *,
DENSE_RANK() OVER(ORDER BY average_delivery DESC) as denserank,
FROM state_average_delivery
ORDER BY average_delivery DESC
)

SELECT
customer_state,
average_delivery as avg_days_fast_delivery
FROM drank
WHERE denserank <= 5;
```


➤ o/p:

JOB INFORMATION		RESULTS	JSON
Row	customer_state	avg_days_fast_delivery	
1	AC	19.76	
2	RO	19.13	
3	AP	18.73	
4	AM	18.61	
5	RR	16.41	

➤ Chart:



➤ Insights & Recommendations For Q5:

- The delivery time varies significantly, indicating potential issues in the logistics and fulfillment process. The company should investigate and address the factors contributing to longer delivery times to enhance customer satisfaction. The company should focus on optimizing the delivery process to ensure timely and reliable deliveries. Improving logistics, inventory management, and shipping methods can help reduce delivery times and enhance customer satisfaction. Regularly monitoring and analyzing delivery performance metrics will enable the company to identify bottlenecks and implement corrective measures. Implementing automated tracking systems and providing real-time updates to customers can improve transparency and manage customer expectations effectively.
- There are significant variations in average freight values across different states. This could be attributed to factors such as distance, transportation infrastructure, logistics efficiency, and competition in the region. States with higher average freight values should be analyzed to identify potential cost optimization opportunities. The company should explore strategies to optimize transportation routes, negotiate better contracts with logistics providers, and streamline operations to reduce freight costs.
- States with lower average freight values may indicate intense competition among logistics providers. The company should assess pricing strategies to ensure competitiveness while maintaining profitability. States with higher average freight values may benefit from investments in transportation infrastructure and logistics networks. The company should consider collaborating with local authorities and logistics partners to improve infrastructure and streamline operations.
- Analyzing freight values can provide insights into customer satisfaction. Higher freight costs could impact customer perception of value, while lower freight costs may contribute to positive customer experiences. The company should continuously monitor customer feedback and align freight pricing strategies with customer expectations.
- There are notable variations in average delivery times across different states. Factors such as distance, transportation infrastructure, logistics operations, and local efficiency can contribute to these variances. States with higher average delivery times should be closely examined to identify potential bottlenecks and inefficiencies in the logistics process. The company should focus on streamlining operations, optimizing transportation routes, and enhancing coordination with logistics partners to reduce delivery times.
- Delivery time plays a crucial role in customer satisfaction. States with lower average delivery times indicate efficient logistics operations that align with customer expectations. The company should strive to meet or exceed delivery time expectations to enhance customer satisfaction. States with longer average delivery times may benefit from infrastructure investments, such as improving transportation networks, optimizing fulfillment centers, and implementing advanced logistics technologies. The company should collaborate with local authorities and invest in infrastructure to enhance logistics efficiency.

- Continuous monitoring and analysis of delivery times can provide valuable insights into logistics performance. The company should leverage data to identify trends, patterns, and areas for improvement, allowing for data-driven decision-making to optimize delivery times. Some states demonstrate a higher level of efficiency in the delivery process, with orders consistently arriving earlier than the estimated date. It indicates effective coordination between the company, logistics partners, and delivery networks.
- The company should analyze the practices employed in these states to identify key factors contributing to faster delivery times. These may include optimized routing, efficient fulfillment processes, proactive logistics management, and effective communication with customers. Insights gained from the top-performing states can be used to improve delivery performance in other regions. The company should share best practices, implement standardized processes, and replicate successful strategies to enhance delivery speed across the entire customer base.
- Proactively updating customers regarding their order status, providing accurate tracking information, and delivering ahead of the estimated date can significantly enhance customer satisfaction. The company should focus on clear and timely communication to manage customer expectations and build trust.

🚦 Q6: Analysis based on sales, freight and delivery time.

❖ Q6.1: Find the month on month no. of orders placed using different payment types.

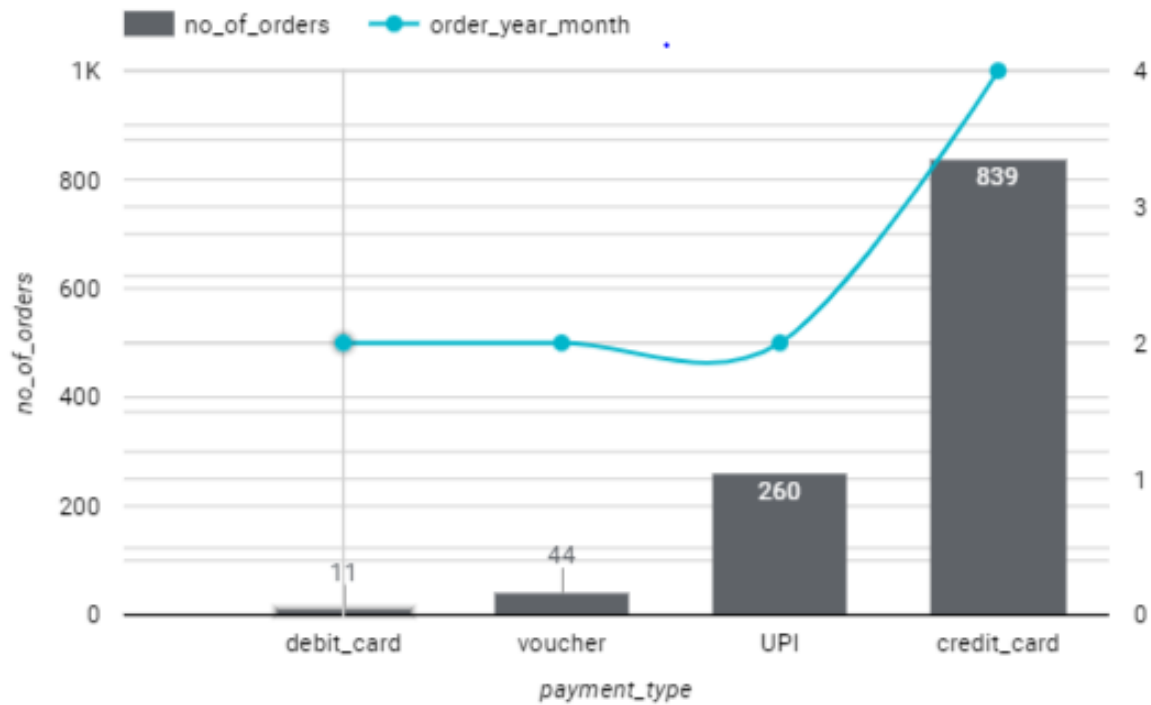
➤ SQL Query:

```
SELECT
FORMAT_TIMESTAMP('%Y-%m', o.order_purchase_timestamp) AS
order_year_month,
p.payment_type,
COUNT(DISTINCT o.order_id) no_of_orders
FROM
`target.customers` c
INNER JOIN `target.orders` o
ON c.customer_id = o.customer_id
INNER JOIN `target.payments` p
On p.order_id = o.order_id
GROUP BY order_year_month, p.payment_type
ORDER BY 1,2,3;
```

➤ o/p:

JOB INFORMATION		RESULTS	JSON
Row	order_year_month	payment_type	no_of_orders
1	2016-09	credit_card	3
2	2016-10	UPI	63
3	2016-10	credit_card	253
4	2016-10	debit_card	2
5	2016-10	voucher	11
6	2016-12	credit_card	1
7	2017-01	UPI	197
8	2017-01	credit_card	582
9	2017-01	debit_card	9
10	2017-01	voucher	33

➤ Chart:



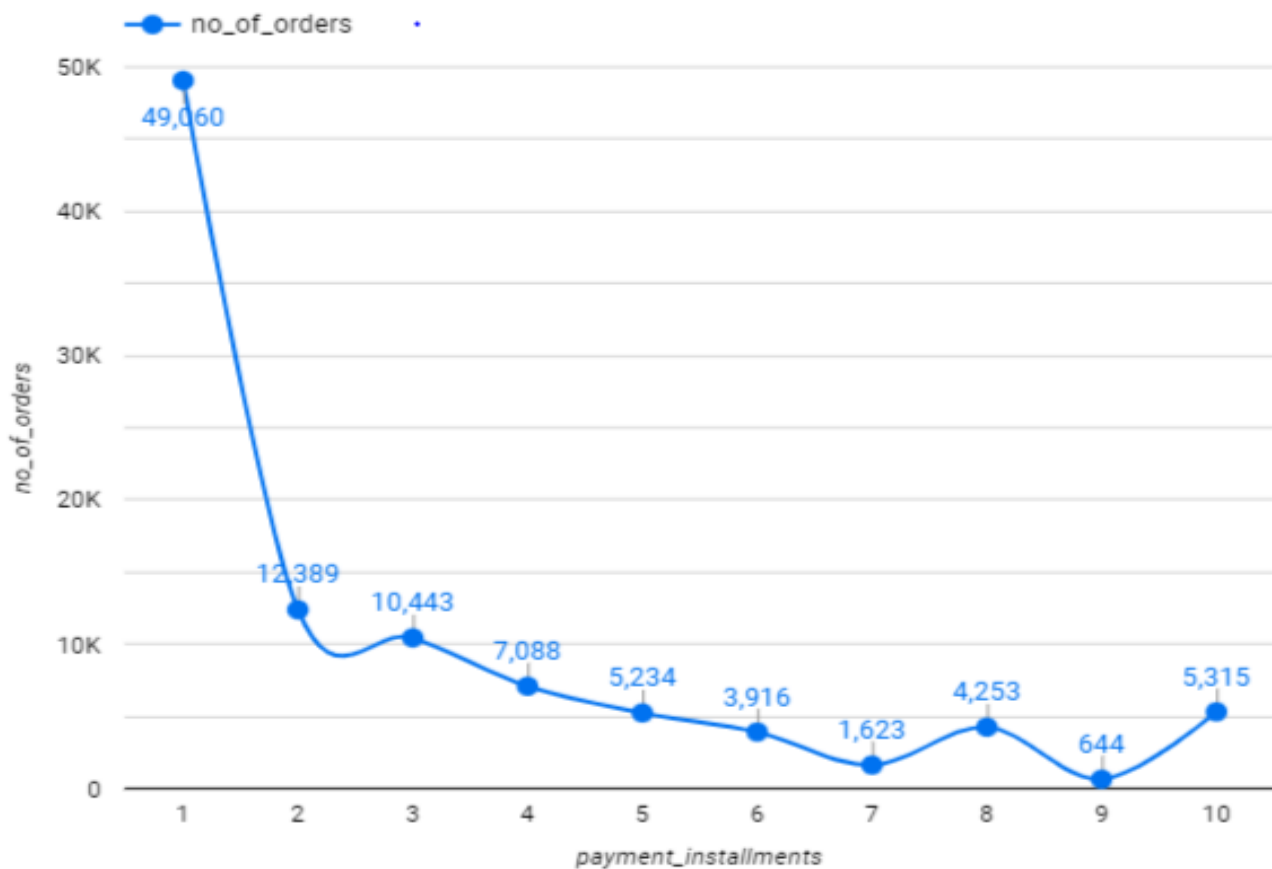
- ❖ **Q6.2:** Find the no. of orders placed on the basis of the payment installments that have been paid.

➤ `SELECT
payment_installments,
COUNT(DISTINCT order_id) no_of_orders
FROM `target.payments`
WHERE payment_installments != 0
GROUP BY payment_installments
ORDER BY 1,2;`

➤ o/p:

JOB INFORMATION		RESULTS	J
Row	payment_installments	no_of_orders	
1	1	49060	
2	2	12389	
3	3	10443	
4	4	7088	
5	5	5234	
6	6	3916	
7	7	1623	
8	8	4253	
9	9	644	
10	10	5315	

➤ Chart:



➤ Insights & Recommendations For Q6:

- Analyzing the month-on-month order count by payment type can help identify any seasonal patterns or fluctuations in customer preferences. The company should focus on understanding these trends to optimize inventory management, allocate resources efficiently, and tailor marketing campaigns accordingly. The data provides insights into the popularity of different payment types among customers. For example, credit card and UPI (Unified Payment Interface) appear to be widely used. The company should ensure a seamless payment experience for these preferred methods and consider expanding payment options to cater to a broader customer base.
- The company can leverage the popularity of certain payment types to drive sales and customer loyalty. Offering exclusive discounts, promotions, or incentives for specific payment methods can encourage customers to choose those options, leading to increased sales and customer satisfaction. Monitoring the adoption rate of emerging payment methods, such as UPI, can help the company stay ahead of the curve in the rapidly evolving digital payment landscape. By providing a smooth and convenient payment experience for these emerging methods, the company can attract tech-savvy customers and gain a competitive edge.
- Analyzing payment preferences can provide valuable insights into customer behavior and preferences. The company can use this information to personalize marketing efforts, improve customer segmentation, and deliver targeted offers that align with customers' preferred payment methods.
- The data provides insights into the popularity of different payment installment plans among customers. The majority of customers opt for a single installment plan, followed by two and three installments. The company should ensure seamless and flexible installment payment options to cater to the preferences of customers and make it easier for them to complete their payments. Analyzing the distribution of orders across different payment installments can help the company understand customer needs and preferences. By offering customized installment plans based on customer preferences and purchase history, the company can enhance customer satisfaction and increase order completion rates.
- The company can leverage customer data related to payment installments to deliver targeted communication and reminders. Sending timely notifications and reminders about pending installments can help reduce the number of abandoned or delayed payments and improve customer experience. The company should focus on streamlining the payment process for different installment plans. Implementing user-friendly interfaces, secure payment gateways, and automated reminders can simplify the payment process and encourage customers to complete their installments on time.
- Offering dedicated customer support for payment-related queries and issues can instill confidence in customers and encourage them to choose installment plans. Providing assistance and guidance throughout the payment journey can help resolve any concerns and increase customer satisfaction.

✓ Nitish Vishwanath Hatapaki

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